

NP- : , ,

2026-06-06

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1 :

(polynomial-time), n $O(n^k)$. tractable(), k $O(n^k)$
intractable() .

“ ”

- (undecidable): (Halting Problem)
- (intractable): k $O(n^k)$, (superpolynomial)

NP- : $\Theta(n^{100})$, tractable .

2

, NP- .

(P)	(NP-complete)
: $O(V E)$	
: $O(E)$.	
2-CNF :	3-CNF

(, 2 3) .

2.1 vs

LONG-PATH (): $G = (V, E)$ k , k ?

2.1.1 NP- —

$|V|$, $|V| - 1$. “ $|V| - 1$ ” “ $|V| - 1$ ”,

. HAM-PATH $\leq_P$ LONG-PATH.

. $G = (V, E)$ G $k = |V| - 1$ ().

($\rightarrow$) G $|V|$ $|V| - 1$ LONG-PATH “ ”.

($\leftarrow$) G $\geq |V| - 1$, $|V| - 1$ $|V| - 1$ $\rightarrow$ $\rightarrow$. ■

HAM-PATH NP- LONG-PATH NP-hard , () NP . NP- .

2.1.2 — “ ”

- : “ v ” , Dijkstra · Bellman-Ford DP/
- : “ v ” . v (). DP “ ”($2^{|V|}$) .
- : , DAG DP . “ + ” .

2.2 vs

Euler tour	Hamiltonian cycle
$O(E)$	$(\quad)$ NP-

: Euler tour, Hamiltonian cycle.

2.2.1 Euler tour

Euler tour : $\forall v : \text{in-degree}(v) = \text{out-degree}(v)$.

$O(V + E)$.

: v , $v \xrightarrow{t}$ in-edge t out-edge t . in-degree = out-degree.

$(\quad, \quad)$:

$(\quad)$. v (out-edge). v in-edge out-edge 1. in-degree(v) = out-degree(v) out-edge.

, (Hierholzer, $O(E)$). “ ”.

2.2.2 Hamiltonian cycle

“ ”. — 3-CNF-SAT NP-.

: Euler tour Hamiltonian cycle.

2.3 2-CNF vs 3-CNF

(satisfiable) 0/1 1. k -CNF k OR (clause) AND. 2-CNF, 3-CNF NP-.

3 : P, NP, NPC

3.1 P

. $O(n^k)$.

3.2 NP

(verifiable). (certificate),.

3.2.1

$\langle v_1, \dots, v_{|V|} \rangle$. : (1) V , (2) $(v_{|V|}, v_1)$.

$O(V)$	$O(V + E)$
$O(V)$	$O(V)$
$O(V ^2)$	$O(V + E)$

- : boolean $O(|V|)$, $O(|V|)$.
- : $O(1)$ $O(|V|)$, $O(|V| + |E|)$.

$O(|V|)$ $O(|V|^2)$ - . , NP . (, NP-)

3.3 P NP

P . $P \subseteq NP$. , $P \stackrel{?}{=} NP$. $P \neq NP$.

3.4 NPC

NP NP . : NP- , NP ($P = NP$).

3.4.1 “ ”

: “NPC ?” NP- , .

- NP- “NP , NP ” , .
- “intractable” ($P \neq NP$) .
- NP-hard (Section 7.1).

: NP , NPC . “ ” . “ ” . “ (NPC)” .

4 P vs NP

2026 P NP . NP- . 2000 100 . $P = NP$ $P \neq NP$. NP- , .

5 vs

() , NP- “ / ” . k .

- SHORTEST-PATH(): $u \ v$.
- PATH(): $u \ v \leq k$?

. SHORTEST-PATH k PATH .

6 (Polynomial-time Reduction)

6.1

$A \alpha \ B \ \beta$.

1. .
2. $\alpha \text{ “ ”} \iff \beta \text{ “ ”}$ (iff).

$A \leq_P B$, “ $A \ B$ ” .

$A \rightarrow B$ (), “ $B \ A$ ” , $A \leq B$.

6.2

- **1** ($\leq$): $A \leq_P B \iff B \in P \iff A \in P$.
- **2** ($\leq$): $A \leq_P B \iff A \text{ NP-hard} \iff B \text{ NP-hard}$.

6.3 (direction)

X .

			X
X	$(X \in P)$	C	$X \leq_P C$
X	$(X \text{ NP-hard})$	A	$A \leq_P X$

NP- . :

(reduce **FROM** known-hard **TO** yours).

6.3.1 ()

VERTEX-COVER NP-hard $\text{CLIQUE} \leq_P \text{VERTEX-COVER}$.

. VC . $\text{CLIQUE} \leq_P \text{VC}$ CLIQUE VC CLIQUE . CLIQUE
 NP- () . . VC . ■
 $(\text{VC} \leq_P \text{CLIQUE})$ “ CLIQUE VC ” , VC . .

6.4

$\leq_P$:

$$X \leq_P Y \wedge Y \leq_P Z \implies X \leq_P Z,$$

(). .

NP- . $A' \in NP$ $A' \leq_P A$, **NP-** $A'' \leq_P A$:

$$A' \leq_P A'' \leq_P A \quad (\forall A' \in NP).$$

6.4.1

$A' \leq_P A'' \leq_P B \leq_P A$ (A'' NPC, A' NP):

- $A' \leq_P A \rightarrow A$ **NP-hard** (B).
- $A' \leq_P A'' \leq_P B \rightarrow A' \leq_P B \rightarrow B$ **NP-hard**.

. anchor A'' (A') “ ” () , anchor (B, A) NP-hard . $\leq_P$.
 , B NP- $B \in NP$. B “ NPC ” — .

6.5

($P \neq NP$). . , .

1. : “ $B \leq_P A$ ” $P = NP$. . . (“ ” “ ” .)
2. **intractability** + : B NP- . . . (“ ” “ ” .)
3. () : (: $\rightarrow$, $\rightarrow$).
4. **NP** : NP- “ ” — , .

7 NP-hard / NP-complete

- **A NP-hard:** $A' \in NP \implies A' \leq_P A$.
- **A NP-complete:** $A \in NP$ and A NP-hard.

“ ” . “hard ” .

7.1 Theorem (34.4)

NP- $P = NP$.

. $L \in P$ $L \in NPC$. $L' \in NP$ NP- $L' \leq_P L$. $L \in P$ 1 $L' \in P$. $NP \subseteq P$, $P \subseteq NP$ $P = NP$. ■

: NP NP- . “NP- = intractability ” .

8

NP- .

1. $X \in NP$:
2. X **NP-hard:** NP- Y $Y \leq_P X$.
 - (a) f + .
 - (b) $(\rightarrow)$: yes yes (yes-).
 - (c) $(\leftarrow)$: yes yes (yes).

8.1 “no no”

$\alpha \in A \iff f(\alpha) \in B$.

$$P \leftrightarrow Q \equiv \neg P \leftrightarrow \neg Q$$

“yes $\leftrightarrow$ yes” “no $\leftrightarrow$ no” ().

, no- :

	yes-	= no-
$(\rightarrow)$	$C \text{ SAT} \Rightarrow \phi \text{ SAT}$	$\phi \text{ UNSAT} \Rightarrow C \text{ UNSAT}$
$(\leftarrow)$	$\phi \text{ SAT} \Rightarrow C \text{ SAT}$	$C \text{ UNSAT} \Rightarrow \phi \text{ UNSAT}$

” $C \text{ UNSAT} \iff \phi \text{ UNSAT}$, no $\leftrightarrow$ no . $(\leftarrow)$ “ $C \text{ UNSAT} \Rightarrow \phi \text{ UNSAT}$ ” “
 . no iff .

9 NP-

NP- .

CIRCUIT-SAT $\rightarrow$ SAT $\rightarrow$ 3-CNF-SAT $\rightarrow$ CLIQUE $\rightarrow$ VERTEX-COVER

9.1 CIRCUIT-SAT — NP-

: AND/OR/NOT (1)?

- **NP:** . , 1 $\rightarrow$.
- **NP-hard (Lemma 34.6):** (NP) .

9.2 SAT — CIRCUIT-SAT $\leq_P$ SAT

: ϕ ?

- **NP:**
- : C x_i x_i , ϕ .

$(\rightarrow) C$ 1. 1 , ϕ 1.

$(\leftarrow) \phi$ 1 1. , C 1 . ■

9.2.1 :

$\rightarrow$ (“ $p \rightarrow q$ ”): $1 \rightarrow 0$, ().

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

: $p \rightarrow q \equiv \neg p \vee q$. “ ” — p .

$\leftrightarrow$ (“ ”): $p \leftrightarrow q$ 1.

• $\phi = ((x_1 \rightarrow x_2) \vee \neg(\neg x_1 \leftrightarrow x_3) \vee x_4) \wedge \neg x_2$, $x_1=0, x_2=0, x_3=1, x_4=1$.

- $x_1 \rightarrow x_2 = 0 \rightarrow 0 = 1$ ().
- $\neg(\neg x_1 \leftrightarrow x_3) = \neg(1 \leftrightarrow 1) = \neg 1 = 0$.
- $x_4 = 1$.
- OR: $1 \vee 0 \vee 1 = 1$.
- $\neg x_2 = 1$.
- $\phi = 1 \wedge 1 = 1 \rightarrow$.

$0 \rightarrow 0 = 1$.

9.3 3-CNF-SAT — SAT $\leq_P$ 3-CNF-SAT (Theorem 34.10)

3-CNF: 3 OR AND. : $(x_1 \vee \neg x_1 \vee \neg x_2) \wedge (x_3 \vee x_2 \vee x_4) \wedge (\neg x_1 \vee \neg x_3 \vee \neg x_4)$.

- **NP:** SAT NP .
- : ϕ 3-CNF ϕ' ($\rightarrow$ 3 $\rightarrow$).

Theorem 34.10. 3-CNF NP- .

9.4 CLIQUE — 3-CNF-SAT $\leq_P$ CLIQUE

: $G = (V, E)$ (). : k ?

- **NP:** V' $\rightarrow$.

• $\phi = C_1 \wedge \dots \wedge C_k$, $C_r = (l_1^r \vee l_2^r \vee l_3^r)$.

- : C_r v_1^r, v_2^r, v_3^r ($3k$).
- : $v_i^r v_j^s \iff (1) (r \neq s) (2) (l_i^r l_j^s)$.

k . $O((3k)^2)$.

$(\rightarrow) \phi$ 1 . $V'(k)$. V' $v_i^r, v_j^s (r \neq s)$ 1 (1),
 $\rightarrow \rightarrow V'$.

$(\leftarrow) G \leq_k V'$, $V' \leq V$. $v_i^r \in V' \leq 1$, $\rightarrow \phi$. ■

: (,) , NP-hard .

9.5 VERTEX-COVER — CLIQUE $\leq_P$ VERTEX-COVER

: $(u, v) \in E \iff u \in V' \wedge v \in V'$. $V' \leq_k V$?

• **NP:** $V' \leq_k V$, $|V'| = k$, $\rightarrow O(|V||E|)$.

. $G = (V, E)$ $\bar{G} = (V, \bar{E})$, $\bar{E} = \{(u, v) : u \neq v, (u, v) \notin E\}$. $(G, k) \leq (\bar{G}, |V| - k)$ ($O(|V|^2)$).

: $G \leq_k V' \iff \bar{G} \leq_{|V|-k} V'$.

$(\rightarrow) G \leq_k V'$. $V' \leq V$. “ $u, v \in V' \Rightarrow (u, v) \in E$.” “ $(u, v) \notin E \Rightarrow u \notin V' \vee v \notin V'$ ”, “ $(u, v) \in \bar{E} \Rightarrow u \in V - V' \vee v \in V - V'$.” $V - V' \leq_{|V|-k} \bar{G}$ ($|V| - k$).

$(\leftarrow) \bar{G} \leq_{|V|-k} V'$. “ $(u, v) \in \bar{E} \Rightarrow u \in V - V' \vee v \in V - V'$.” “ $u \notin V' \vee v \notin V' \Rightarrow (u, v) \notin E$, $(u, v) \in E$.” $V - V' \leq_k G \rightarrow V - V' \leq_k G$ (k). ■

10 :

NP-

1. $X \in NP$ — + .
2. X NP-hard — NPC $Y \leq_P X$.
 - (a) , (b) $(\rightarrow)$, (c) $(\leftarrow)$.
3. () NP-hard .

- : known-hard $\rightarrow$ yours (yours $\leq_P$).
- : yours $\rightarrow$ known-easy (yours $\leq_P$).
- $\leq_P$, .

- $p \rightarrow q \equiv \neg p \vee q$ ().
- $P \leftrightarrow Q \equiv \neg P \leftrightarrow \neg Q$ (yes yes no no).
- $(P \rightarrow Q) \equiv (\neg Q \rightarrow \neg P)$ ().